# KEVIN KWAN

Software Engineer II | Conversational AI & Voice Automation
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## SUMMARY

Software Engineer II at GEICO — conversational AI and voice automation. Build and operate the Sierra AI billing voice agent and Amazon Connect IVR on AWS: 436,000+ conversations at ~51% containment, ~30,000 calls/week.

## WORK EXPERIENCE

**GEICO** July 2024 – Present
*Atlanta, GA (Remote)*
**Software Engineer II — Conversational AI & IVR** February 2026 – Present
**Software Engineer I — Technology Development Program** July 2024 – February 2026

- Spearheaded the Sierra AI billing voice agent (live since May 2026) as its largest contributor, containing ~51% of 436,000+ English and Spanish conversations without live-agent transfer (May–Oct 2026) and improving containment from 48% to 52% as volume scaled ~10x to ~30,000 calls/week across payments, rescheduling, AutoPay, and billing questions.
- Delivered end-to-end make-payment flows for the agent, processing 22,600+ payments at 92–95% authorization success across saved methods, pay-now or scheduled payments, debit authorization, declined or expired card recovery, and policies pending cancellation; shipped net-new Pause AutoPay, 82,800 reschedules, and 192,000 informational answers at 98% tool success.
- Re-architected the billing agent from regex- and classifier-based routing to fact-based journeys (~20,000 net lines removed) and built simulation-based regression suites and automated code-review quality gates for high-velocity releases.
- Resolved 86 production and pre-release billing bugs at the root cause; reviewed code for 12+ engineers and onboarded two engineers onto the Sierra platform.
- Owned production reliability across release validation, live IVR testing, outage rollbacks, and postmortems, and led a full-volume conversation-experience review to prioritize and resolve top customer-experience gaps.
- Migrated GEICO's legacy Cisco UCCE and Nuance IVR to Amazon Connect on AWS, engineering Java traffic-redirection logic for gradual live caller cutover alongside legacy endpoint and Amazon Lex pattern migrations.
- Led the Proof of Coverage IVR (~80% containment) and its disaster-recovery rebuild pipeline, and built the AWS self-service payment IVR (~60% containment) with a zero-amount-due flow, automated declined-payment retries, and new card tokenization.
- Engineered high-throughput, asynchronous voice-workflow orchestration using AWS Lambda, API Gateway, DynamoDB, and Amazon Lex (JavaScript code hooks) across Amazon Connect contact flows, securing customer PII and payment APIs with AES-256 encryption and automating infrastructure deployments via Terraform.

**Publicis Sapient (Publicis Groupe)** June 2023 – August 2023
*Software Engineer Intern | Atlanta, GA*

- Built a WhatsApp AI chatbot for the Travel & Hospitality industry using Google Vertex AI's PaLM LLM, LangChain, MongoDB, and external APIs to deliver live data and recommendations for Atlanta attractions, hotels, and restaurants, achieving 91% response accuracy.
- Developed Python/Flask RESTful APIs bridging MongoDB, the AI chatbot backend, and an interactive map website, automating CI/CD pipelines via GitLab and Docker to deploy on Google App Engine and Firebase Hosting.

**Georgia Tech EcoCAR** January 2022 – May 2023
*Software Engineer, Connected & Automated Vehicles (V2X Subteam) | EcoCAR (Electric Vehicle) Mobility Challenge | Atlanta, GA*

- Designed a camera/sensor state machine for automated stop/go decisions across traffic-light scenarios, improving fuel efficiency and ride experience to help Georgia Tech secure 1st place in the U.S. Department of Energy's 2022 EcoCAR Mobility Challenge.

## TECHNICAL SKILLS

**Languages:** Java, Python, TypeScript, JavaScript, C#, C++
**Conversational AI & Voice:** Sierra AI, Amazon Connect, Amazon Lex, LangChain, Google Vertex AI, OpenAI, Cyara, LLM Evaluation, Agentic Workflows, Tool Calling
**AWS & Cloud Infrastructure:** AWS (Lambda, DynamoDB, API Gateway, S3, CloudWatch), Serverless, Event-Driven Systems, Google Cloud Platform (App Engine, Firebase), Microsoft Azure, Cloudflare
**Web, APIs & Data:** REST APIs, Flask, Node.js, React, Next.js, Tailwind CSS, MySQL, MongoDB, Cloud Firestore
**DevOps & Tooling:** Terraform, Docker, CI/CD, Git, GitHub, GitLab, SonarQube, Cursor, Claude Code

## EDUCATION

**Georgia Institute of Technology — Atlanta, GA** August 2021 – May 2024

- Bachelor of Science in Computer Science, College of Computing; Threads: Intelligence and Media; *summa cum laude* GPA: 3.9
- Zell Miller Scholarship, Graduated with Highest Honors, Dean's List, Faculty Honors